Data4All Guest Lecture

Primer on the Development of Large Language Models



Understanding the Development Process Behind LLMs

Grace Li | October 31, 2026

Introduction



Who am I?

- I'm Grace, a current Computer Science PhD Student at the University of Chicago

- Before the PhD, I double majored in English and Computer Science at Columbia University in NYC.

- Right now, I'm conducting research on Large Language Models (LLMs) and how we can help students become more informed users of this technology.

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*I've done AI research at **Morgan Stanley** on the Investment Banking Division (IBD) and at **Microsoft Research** on their Fairness, Accountability, Transparency, and Ethics Team (FATE).*

Participate in research!

Study Title: Exploring AI Literacy Effects on Student Interaction Behaviors with Large Language Models

This is your chance to help inform the future of AI literacy. By consenting to research, your experiences will help us understand how students learn about AI and to improve educational materials to teach AI literacy.

What does this mean for you? All students will get the same experience today. If you and your parent have consented to participate, after the workshop, we will use your work from today as part of our research project on how students learn AI.

If you are interested, please sign the student and parent consent forms.

Today's Agenda

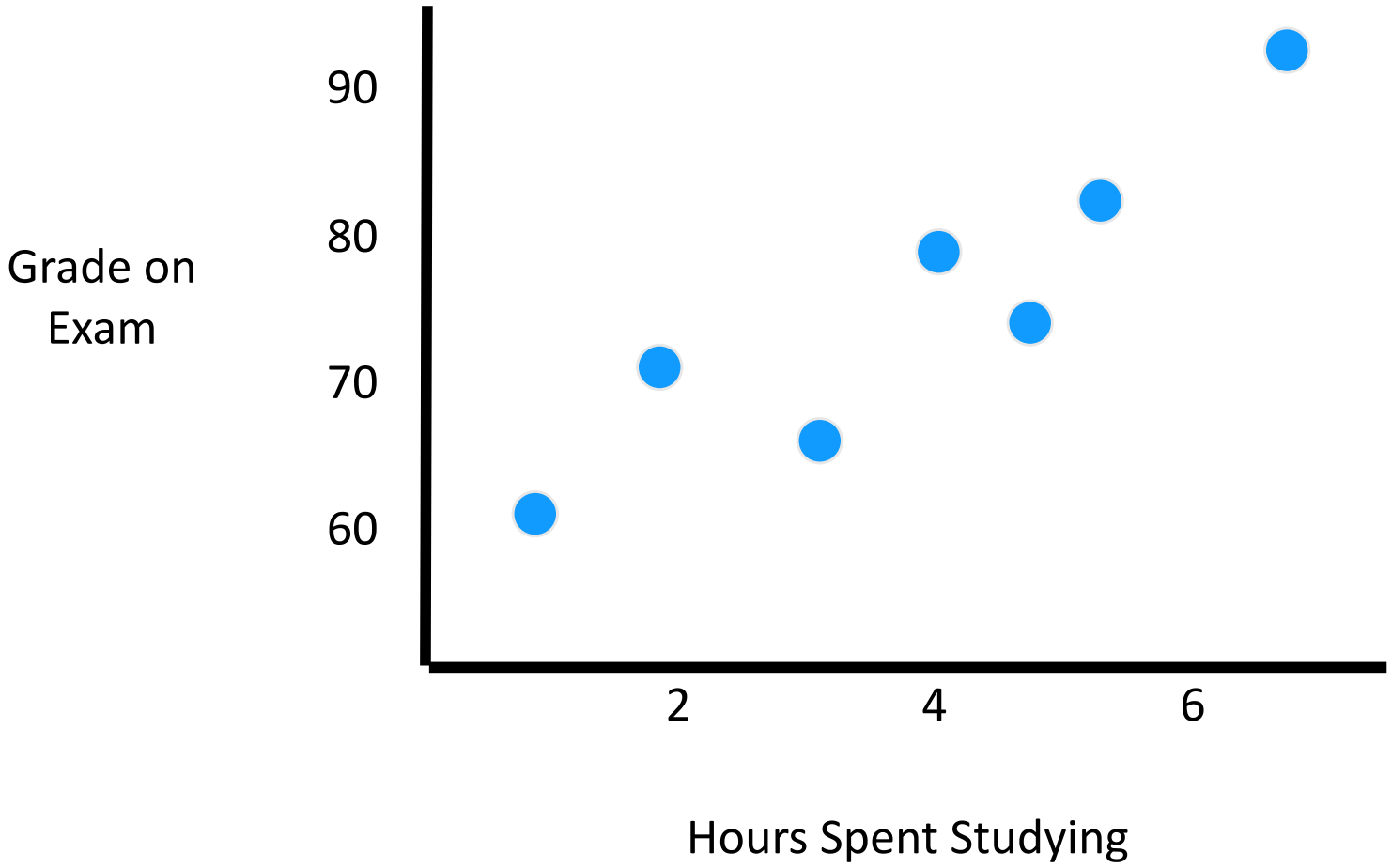
1. How does generative AI relate to data science concepts?

2. How are generative AI chatbots like ChatGPT developed?

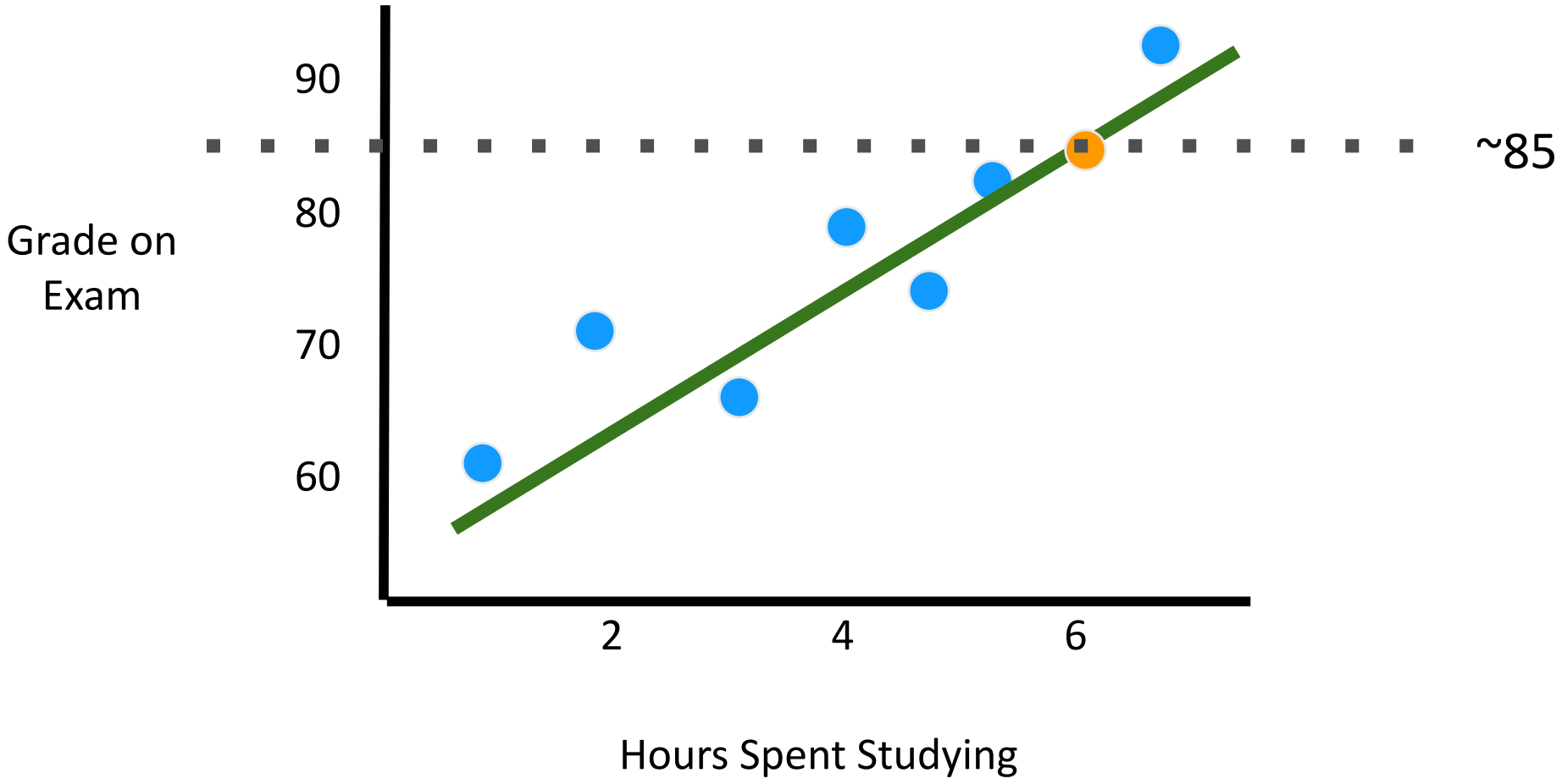
3. Why is understanding the development process of AI chatbots important?
    a. What can it help us understand about AI's societal impacts?

If you have any questions, please feel free to ask at any time throughout the presentation!

What is the predicted exam grade for someone who studies for 6 hours?



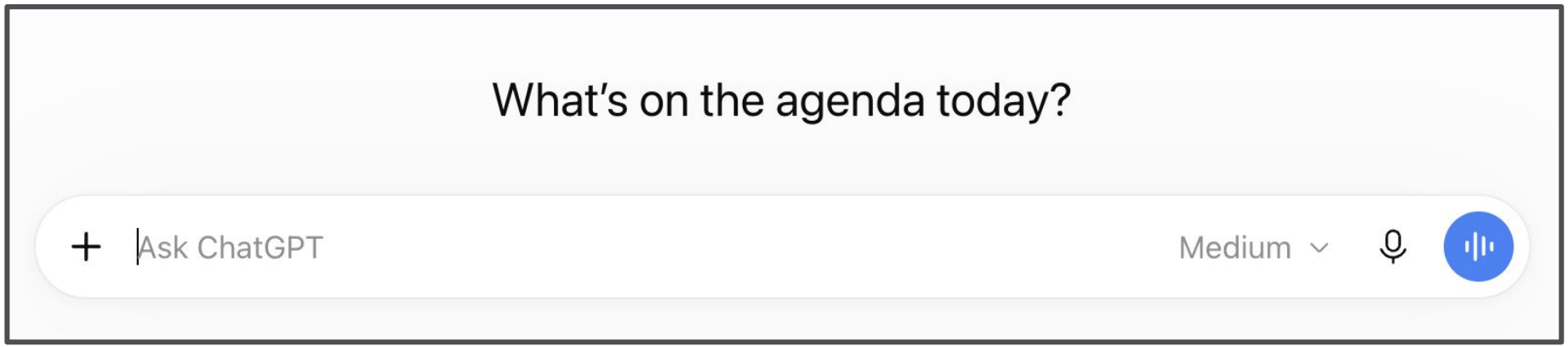
What is the predicted exam grade for someone who studies for 6 hours?



Today's Focus: Large Language Models

Large language models (LLMs) is a form of generative AI that are trained to generate text.

- **Input**: text
- **Output:** text



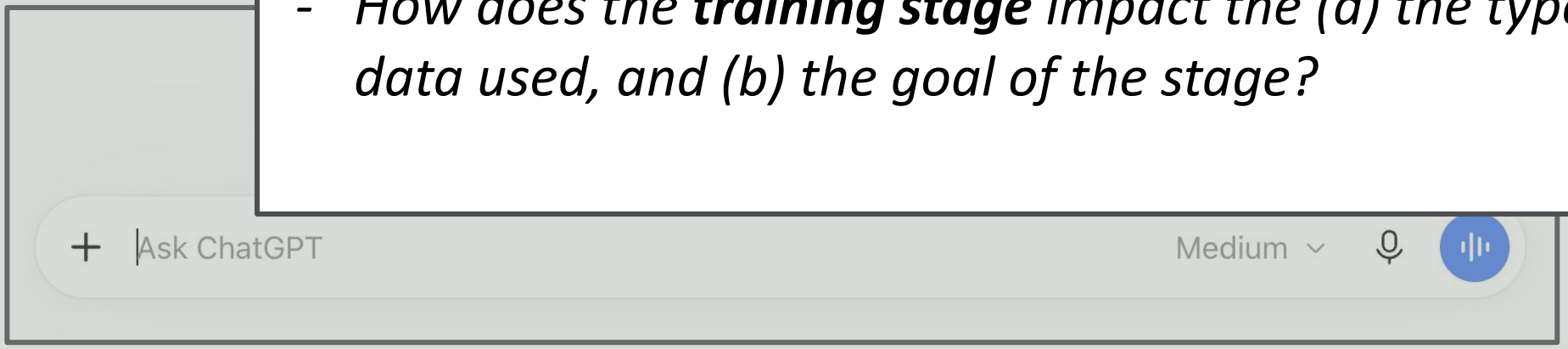
Today's Focus: Large Language Models

Large language models (LLMs) is a form of generative AI that are train to generate te

- **Input**: t
- **Output**:

Guiding Question:

- *What are the **training stages** behind developing large language models?*
- *How does the **training stage** impact the (a) the types of data used, and (b) the goal of the stage?*



Overview of the LLM Training Stages

Step 1: Pre-Training

Goal: _____

Data Used During Training:

_____

Step 2: Post-Training

Goal: _____

Data Used During Training:

_____

Step 3: Deployment

Goal: _____

Data Used During Training:

_____

Worksheet Response:

Make your best guess.

- *What do you think is the goal of each stage of the LLM development process?*
- *What types of data do you think are used to train the model at each stage?*

Hands on Activity:

Can you spot the differences between outputs from a pre-trained, post-trained, and deployed LLM?

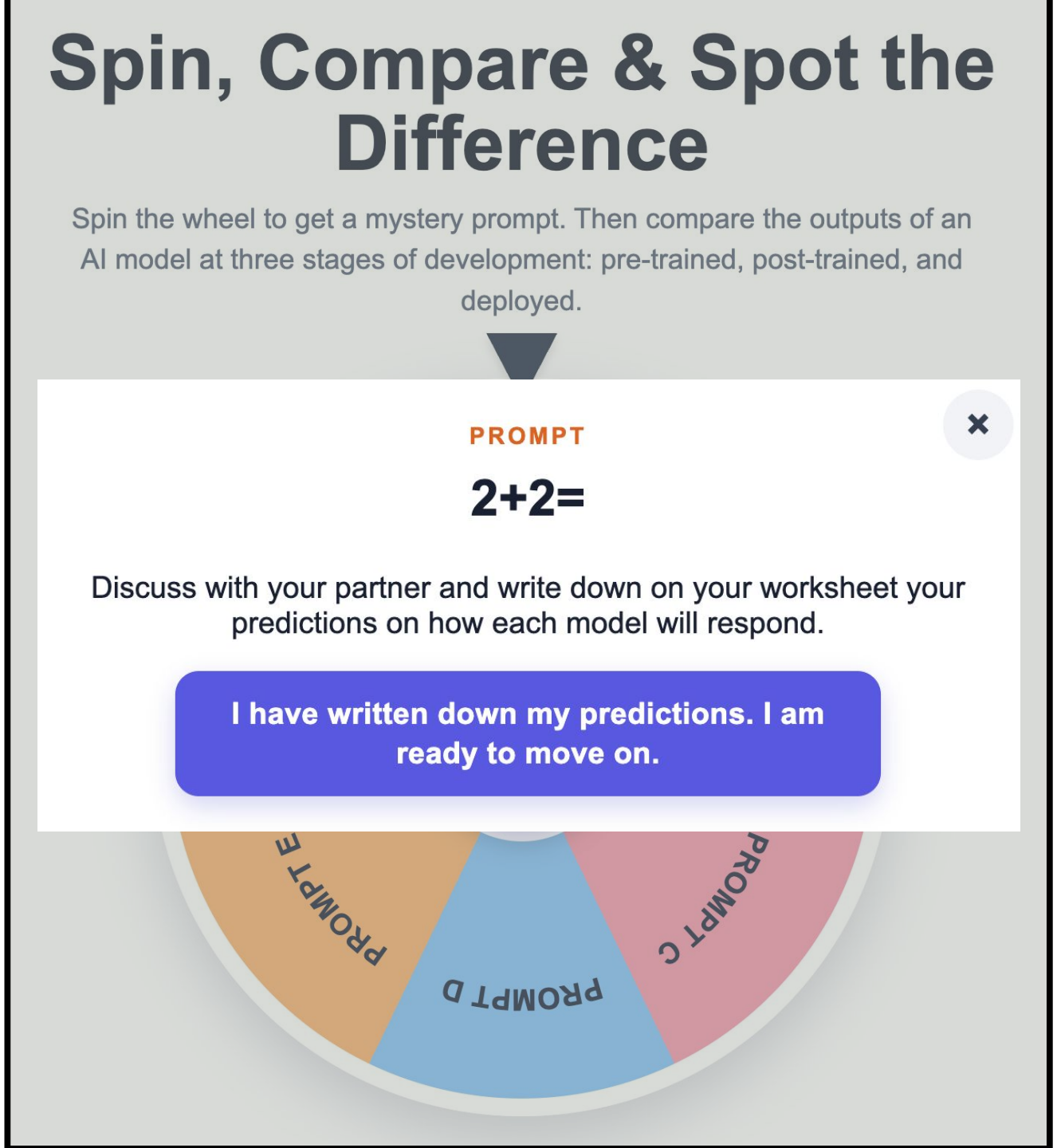
AI Detective Activity

In this activity, you will explore how models at different stages of the LLM development process respond to the same prompt.

1. Spin the wheel for a prompt.

2. Fill in the worksheet for your predictions on how each of the models might respond.

3. Click the button “...I am ready to move on.” To see the outputs.



Compare Pre-trained, Post-trained, and Deployed Models

ORIGINAL PROMPT

2+2=

Generation 1

Pre-trained model

2+2= 4
3+3=6

Post-trained model

2+2=

2+2=4

Deployed model

2+2=

2 + 2 = 4

📝 **Talk with your partner. What differences do you notice between the different models? Write down your observations on the worksheet provided.**

Some questions that you can consider are:

- How is the model output formatted?
- What does each model seem to be able to do?
- Are there similarities that you notice between the different models? Are there differences?

⟳ **Generate another example**

↻ **Respin the wheel**

AI Detective Activity

At your small tables form groups of 2-3 students. Assign these roles to the people in your group:

1. **Operator:** This person will navigate the website and be in charge of clicking the buttons.
2. **Note-Taker:** This person will be in charge of completing the group worksheet with the observations and notes that occurred during the discussion.
3. **Spokesperson** (if 3 person groups): Will be in charge of sharing at least one observation during the large group discussion afterwards.

Operator: Open a new browser tab and search **bit.ly/model-comparision**

Note-taker: Make sure you have a worksheet packet and a pen for your group.

Any questions?

What are some things that you notice about the different models?

Pre-Trained	Post-Trained	Deployed
- asks you questions and doesn't give answers - keep asking the same question, repeating the same word, might repeat the same word multiple times - both pre- and post-trained made up information (made up info about a fake doctor / made up background) - gives very basic factual information, sometimes accurate, sometimes not - loosk like it was copied and pasted from random sources	- concise, but doesn't guarantee veracity of the response - - just gives you the answer, but doesn't explain the answer	- searches the web and doesn't give you what you want - doesn't make up a fake response - provides different websites or resources depending on what you ask - always an emoji in reasoning portion of the response - the model questions about whether the question being asked is (e.g., fake doctors note model will not do it) - sometimes might give wrong info
- past data from encoylpedias, websites, things that are online - outdated info - uses prompts that were given to it	- got some data from conversations with humans (seemed friendlier, ask questions in response to what you are saying) - outdated info - prompts that were given to it at an earlier time	- getting current data and data from humans (/current information) - the only one that can go on the web and search for information - most consistent in relaying that it cannot answer a question

Based on what you observed, what you think are the types of data used to train each model at each stage of the process?

Pre-Trained	Post-Trained	Deployed

Step 1: Pre-Training

2+2= 4

3+3=6

Doesn't just answer the question, but continues the pattern of the input.

What is 4,187 × 36? What is 4,187 × 36? What is 4,187 × 37? What is 4,187 × 38? What is 4,187 × 39? What is 4,187 × 40?

Sometimes the generations might be nonsensical

How many letters are in skibidi? I don't know what skibidi means. Is it a word? It sounds like a made-up word to me.

But, can also generate outputs that seem reasonable and comprehensible.

The main goal of pre-training is to **predict the next word that will come next.**

What words do you think will come next in this sequence?

Who ? too many possibilities ?

Who lives in a house? tree? a noun? maybe an adjective

Who lives in a pineapple under the sea? Spongebob Squarepants *.*

Why does predicting the next word get easier?

What words do you think will come next in this sequence?

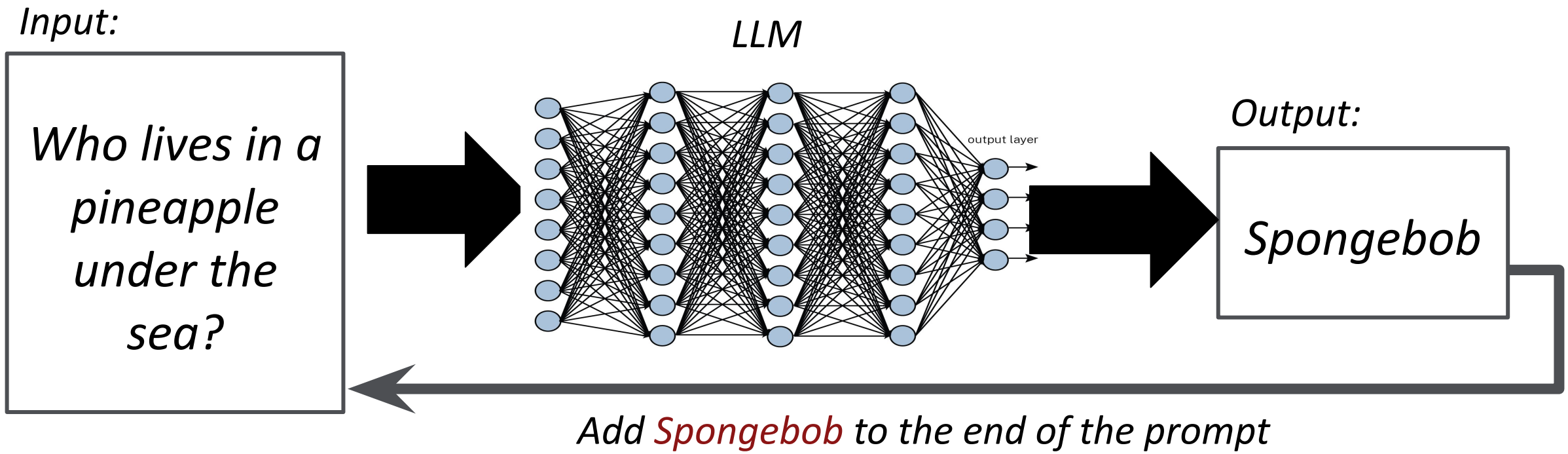
Who ? too many possibilities ?

Who lives in a house? tree? a noun? maybe an adjective

Who lives in a pineapple under the sea? Spongebob

Context Clues provide hints as the types of words that are most likely to come next.

How does next word prediction look like for the LLM?



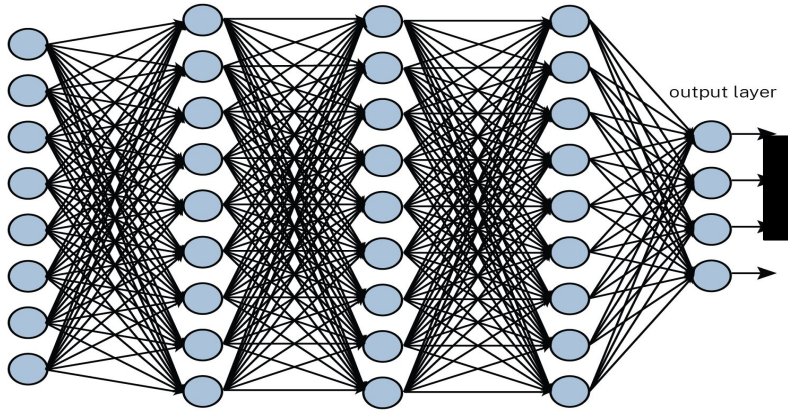
But LLMs typically generate more than just one word! How does that happen?

How does next word prediction look like for the LLM?

Input:

Who lives in a pineapple under the sea?
Spongebob

LLM

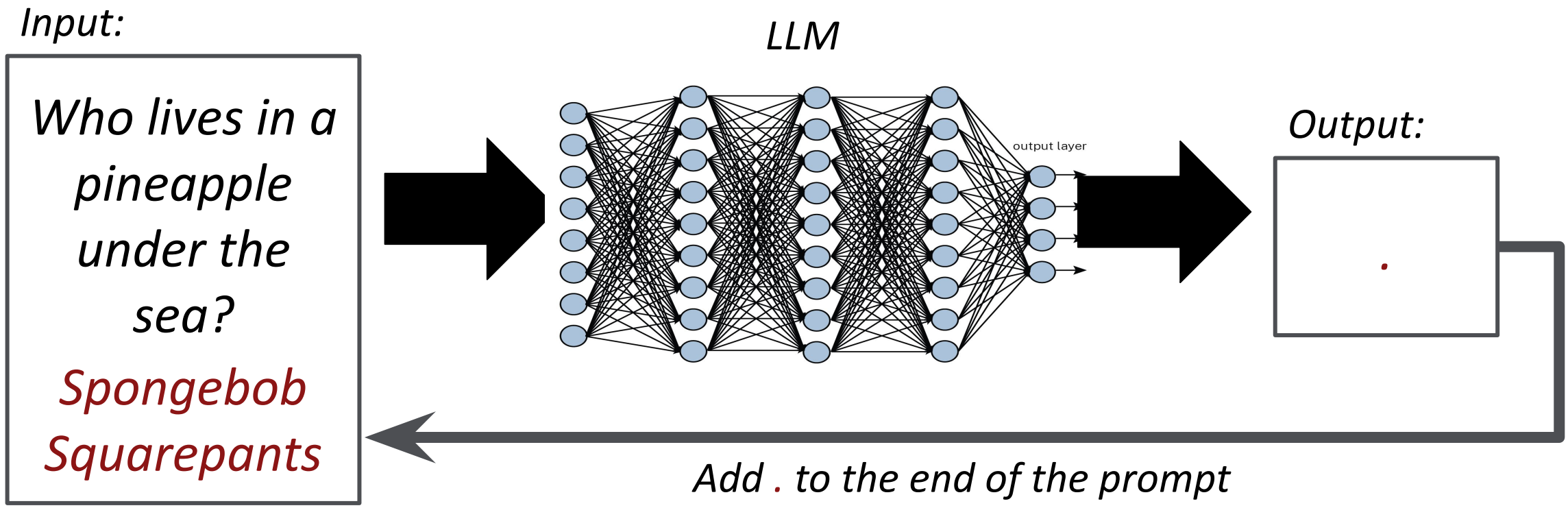


Output:

Squarepants

Add Squarepants to the end of the prompt

How does next word prediction look like for the LLM?



Once a certain amount of words have been generated or a certain ending marker (e.g., period, question mark) has been reached. The generation will stop.

What types of data should we collect to train a model to predict the next word?

Step 1: Pre-Training

Goal: Training a language model to predict the next word

Training Data: Gather all the text that is posted on the internet (Wikipedia, Blog Posts, Online Books, Reddit threads, etc.)

Why? To provide as many examples of the structure of the English language.

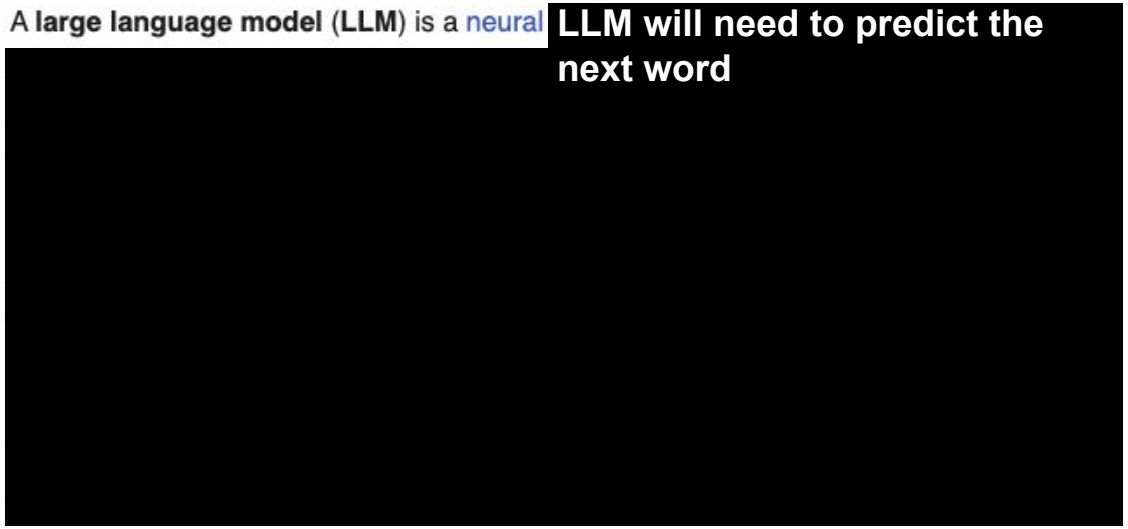
Original Text

A **large language model** (**LLM**) is a neural network trained on a vast amount of text for natural language processing tasks, especially language generation. LLMs can typically generate, summarize, translate, and analyze text in many contexts, and are a foundational technology behind modern chatbots.[1] Biased or inaccurate training data can make an LLM's output less reliable.[2]

LLMs are typically based on transformer architecture.[3] Generative pre-trained transformers (GPTs) are a type of LLM that is pre-trained to predict the next word.[4] GPTs are then often fine-tuned to follow instructions and to behave as assistants.[5]

Benchmark evaluations for LLMs attempt to measure model reasoning, factual accuracy, alignment, and safety.[6]

Training Version of the Text



What happens during the pre-training process?

Original Text

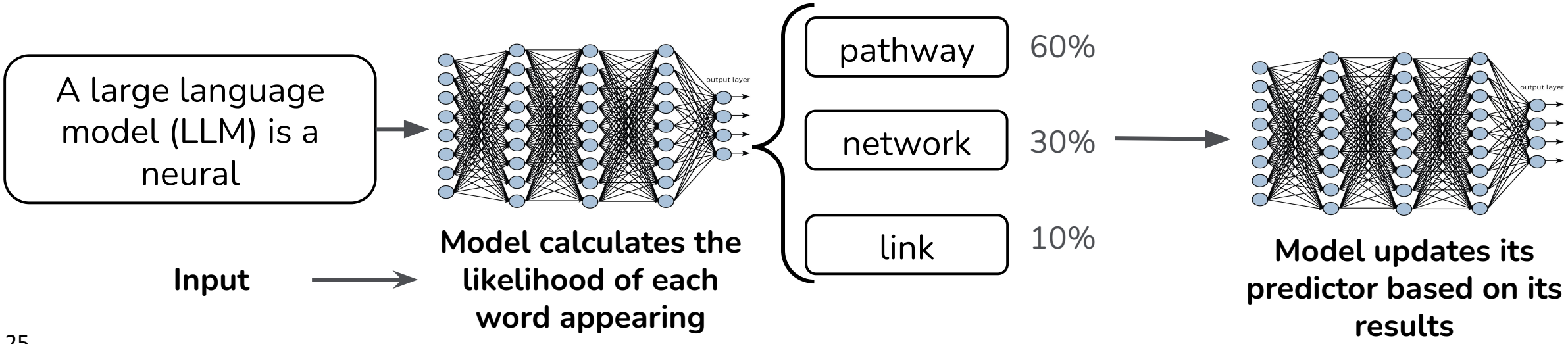
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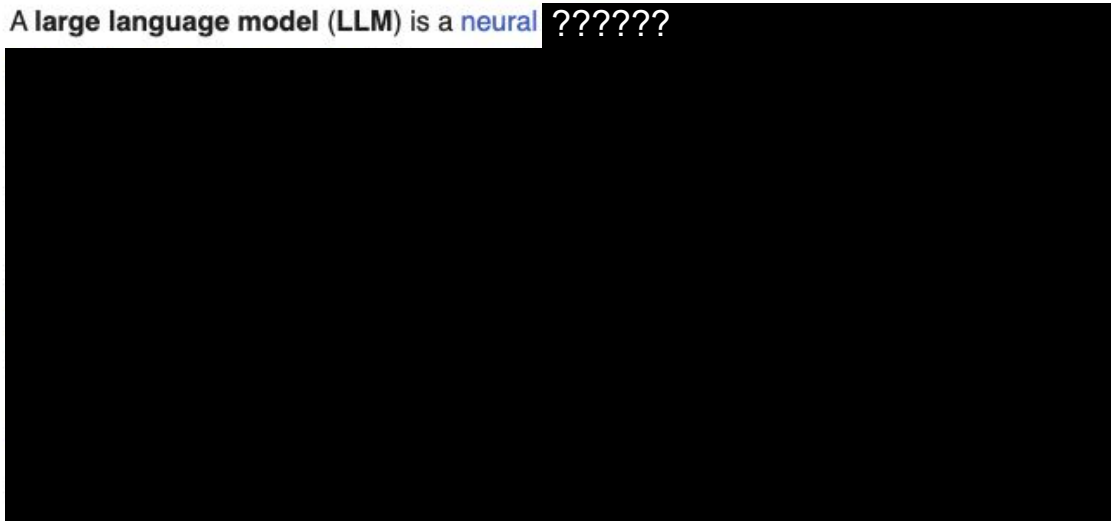
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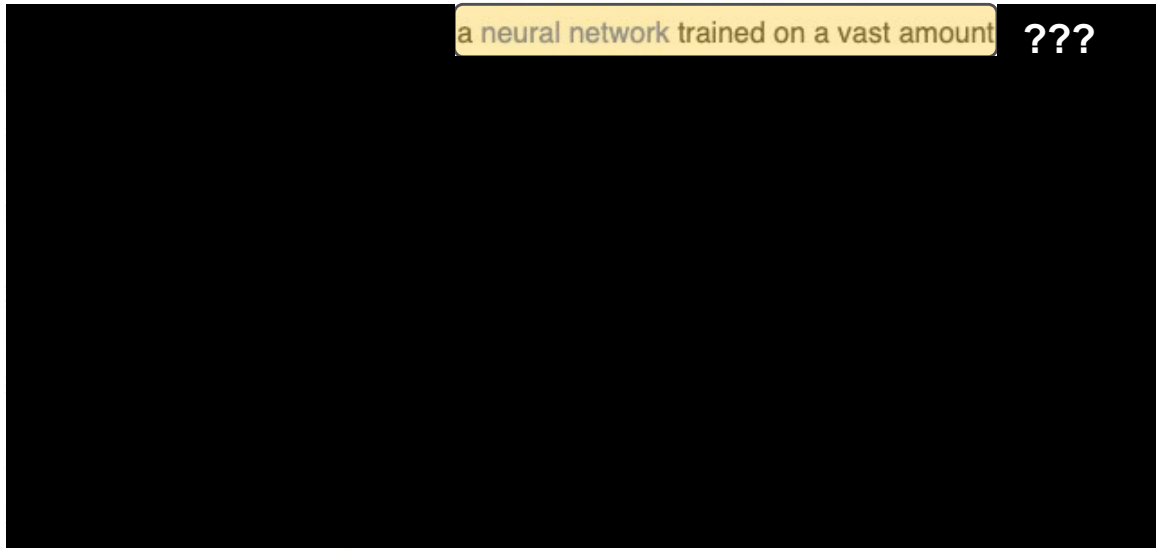
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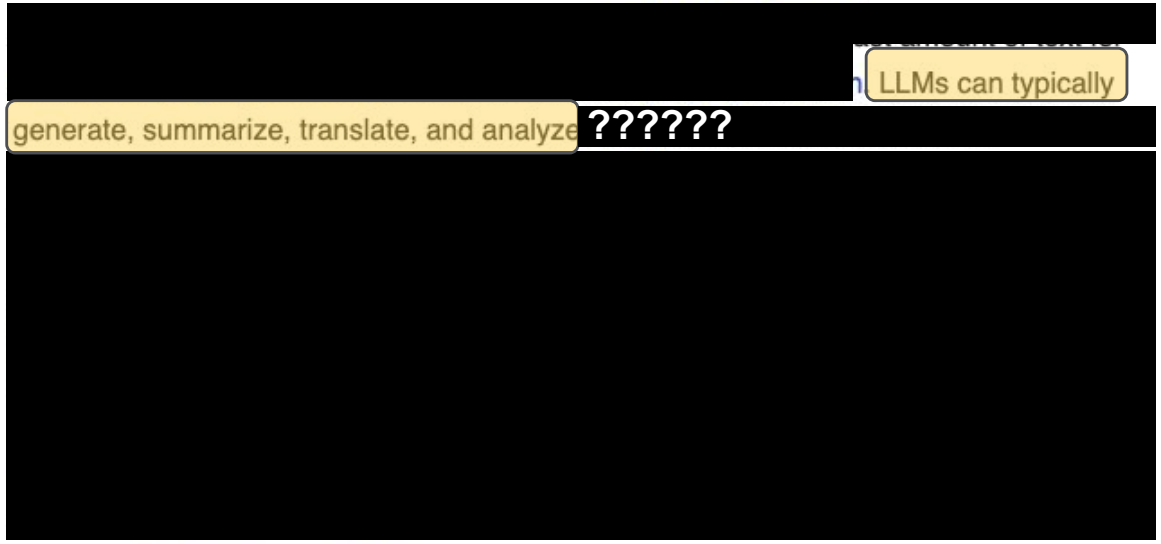
Benchmark evaluations for LLMs attempt to measure model reasoning, factual accuracy, alignment, and safety.[6]

Training Version of the Text



Generating new training examples from the same text sample by varying the **context**.





How does the model perform on various amount of training data?

No training data

"Sunny ? the day purple cardigan , dream because road pavement house; a"

Model produces nonsensical outputs

Trained with 100k examples.

"Sunny day is warm. Day is warm. Warm day is warm. Warm day. Warm day."

Model begins to replicate some basic structures like punctuation and sentence structure.

Trained on 1 trillion examples.

"A sunny day lets children play by the seaside without worrying about catching a cold."

Model begins to replicate more complex sentence structures.

With more training data, the model can generate different sentence structures and patterns based on a diverse range of text it was trained with.

Step 1: Pre-Training Review

Goal: Training a language model to predict the next word.

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But, can also generate outputs that seem reasonable and comprehensible.

*How does an pre-trained model consistently respond in **conversational manner**, like ChatGPT?*

Overview of LLM's Development Process

Goal: Gain an understanding of the different steps of the LLM development process.

Step 1: Pre-Training

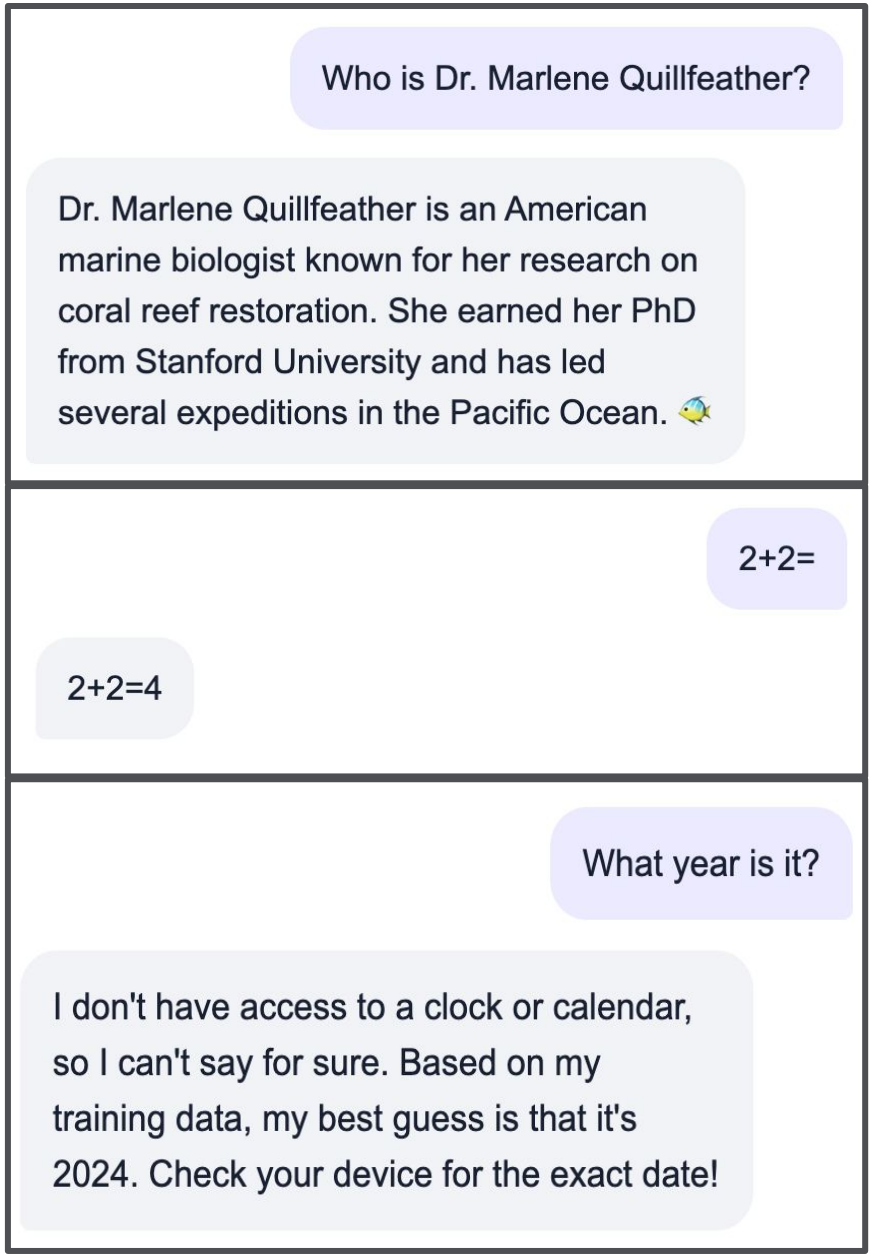
Goal:

- Training a language model to **predict the next word**

Dataset: Using **large scale** text data (aka all the text data internet).

- Wikipedia, blogs, reddit

Step 2: Post-Training



Structure of the LLM outputs are more **"conversational"** like text-messages.

Directly answers the question.

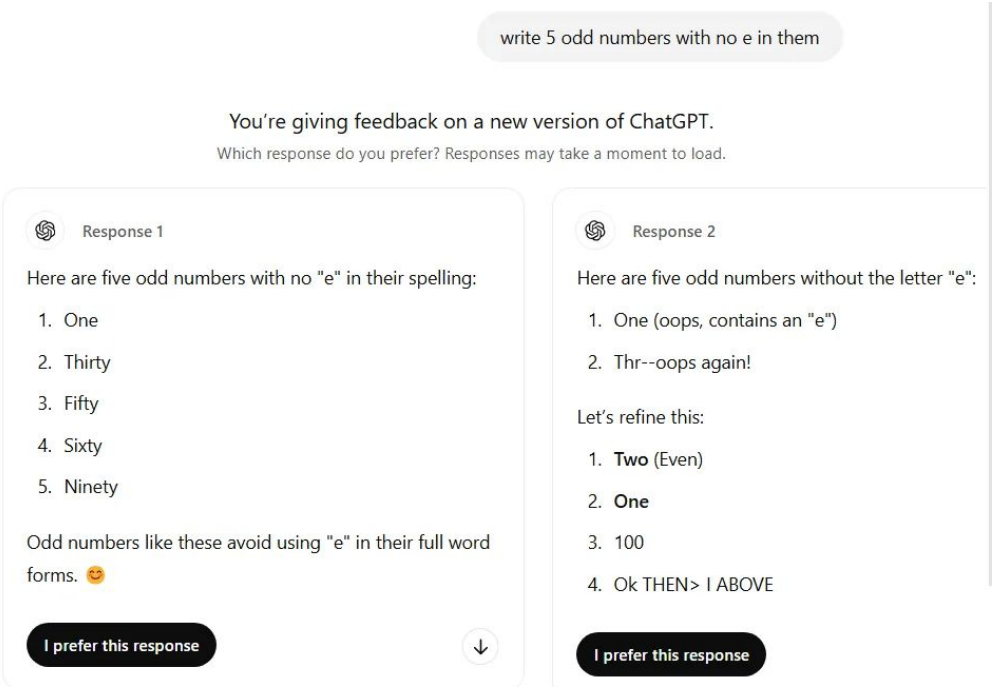
But, there are still missing features that deployed models like ChatGPT like web search.

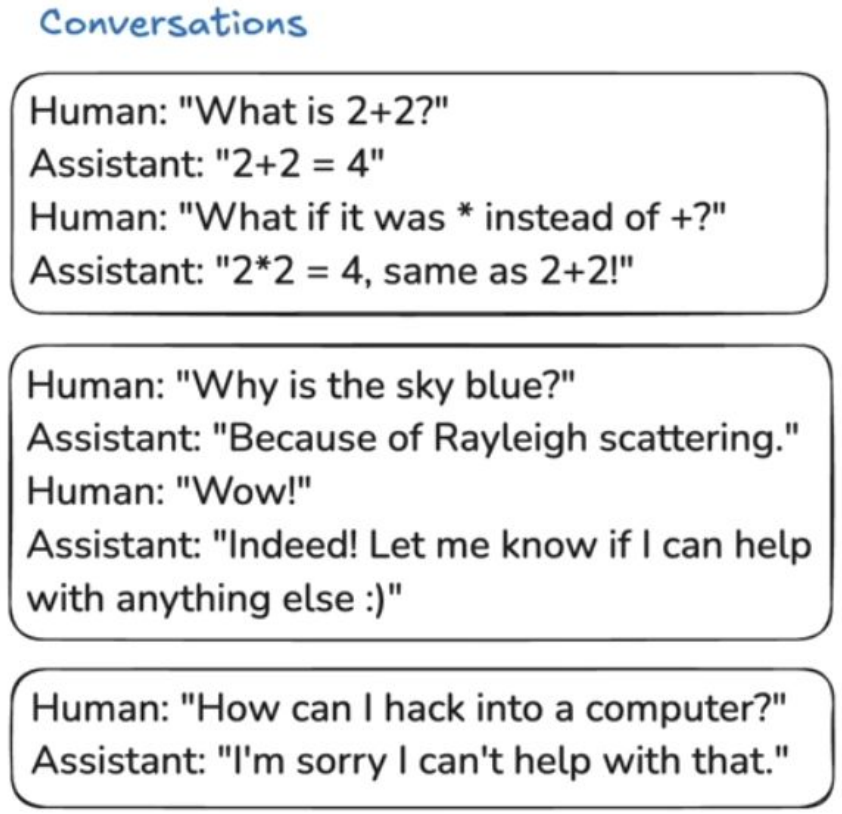
Step 2: Post-Training

Goal: Turn a next-word prediction model into a model that can be a “helpful, friendly, conversational assistant”

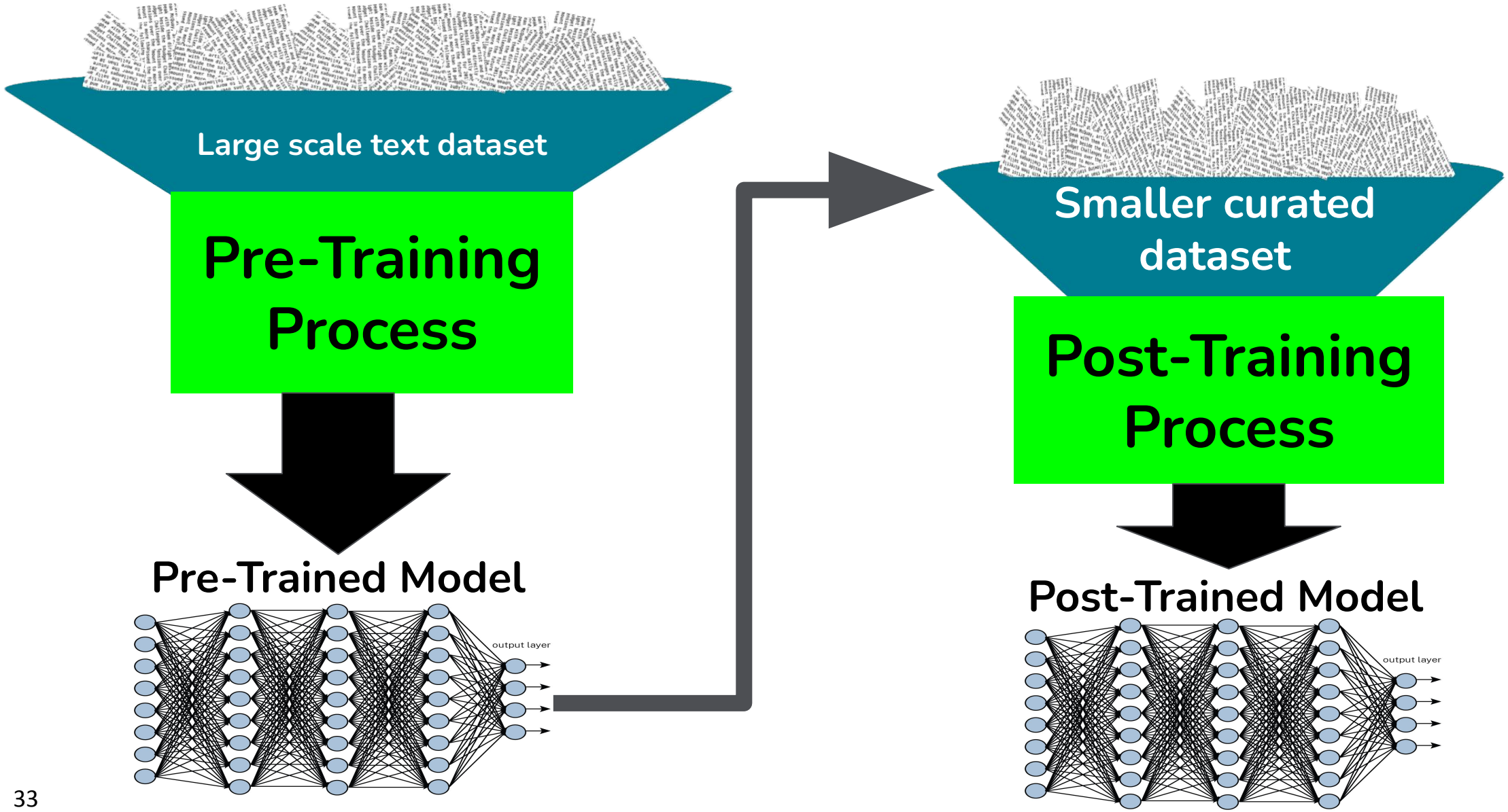
Training Data: Carefully select datasets that are designed to train a model to complete a given task

- Conversational data (e.g., “User:... Assistant:...”)
- Human Preferences





Step 2: Pre-Training and Post-Training Relationship



What does the post-training process look like?

Goal: Turn a next-word prediction model into a model that can be a "helpful, friendly, conversational assistant"

How can we **structure** conversational data as a next word prediction problem?
Only provide the user's prompt and have the model generate the "assistant" output.

Conversations

Human: "What is 2+2?"
Assistant: "2+2 = 4"
Human: "What if it was * instead of +?"
Assistant: "2*2 = 4, same as 2+2!"

Original Data

Human: "What is 2+2?"
Assistant: "2+2 = 4"
Human: "What if it was * instead of +?"
Assistant: "2*2 = 4, same as 2+2!"

Training Data

Human: "What is 2+2?"
Assistant: [redacted]

What does the post-training process look like?

Goal: Turn a next-word prediction model into a model that can be a "helpful, friendly, conversational assistant"

How can
next wor
Only pro
model ge

Conversations

Human: "What is 2+2?"

+?"

What about conversations where the user + assistant go back and forth multiple times?

Original Data

Human: "What is 2+2?"
Assistant: "2+2 = 4"
Human: "What if it was * instead of +?"
Assistant: "2*2 = 4, same as 2+2!"

Training Data

Human: "What is 2+2?"
Assistant: [redacted]

What does the post-training process look like?

Original Data

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Training Data

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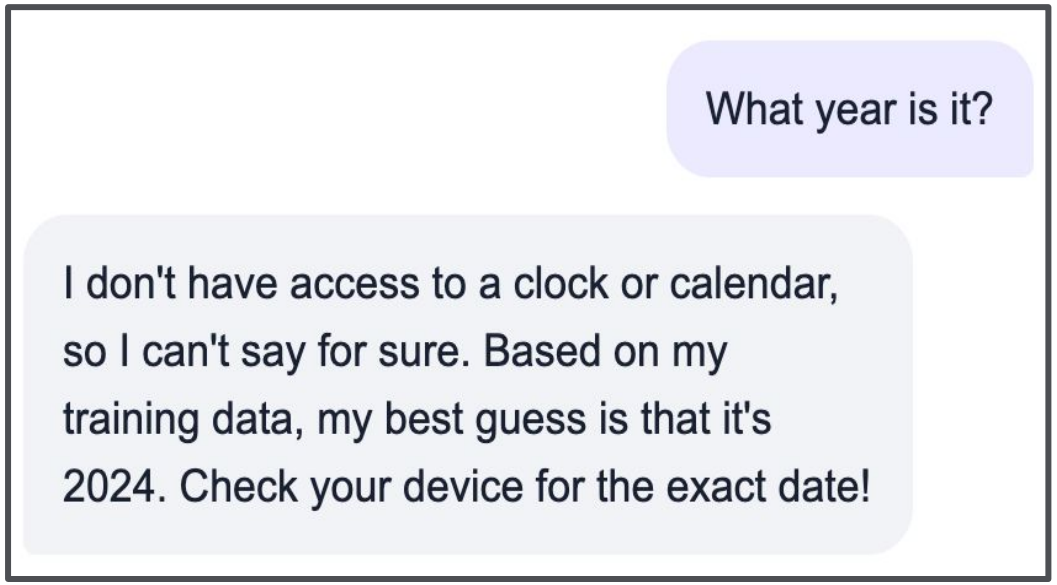
Provide the human message to the model and have the model generate the assistant response. For multiple interactions, show the conversation up until the latest human message, then have the model generate the assistant response.

Step 2: Post-Training Review

Goal: Take a pre-trained model and further **specialize** the next word prediction behavior to train a model to act as a “helpful friendly assistant”

Training Data: Using a smaller and more curated dataset of conversation pairs (e.g., like text messages)

Why? To provide as many examples of the **structure** assistant responses, **tone** assistant should have, and the **style** of conversation.



When are additional features that ChatGPT have added?

Overview of LLM's Development Process

Goal: Gain an understanding of the different steps of the LLM development process.

Step 1: Pre-Training

Goal:

- Training a language model to **predict the next word**

Dataset: Using **large scale** text data (aka all the text data internet).

- Wikipedia, blogs, reddit

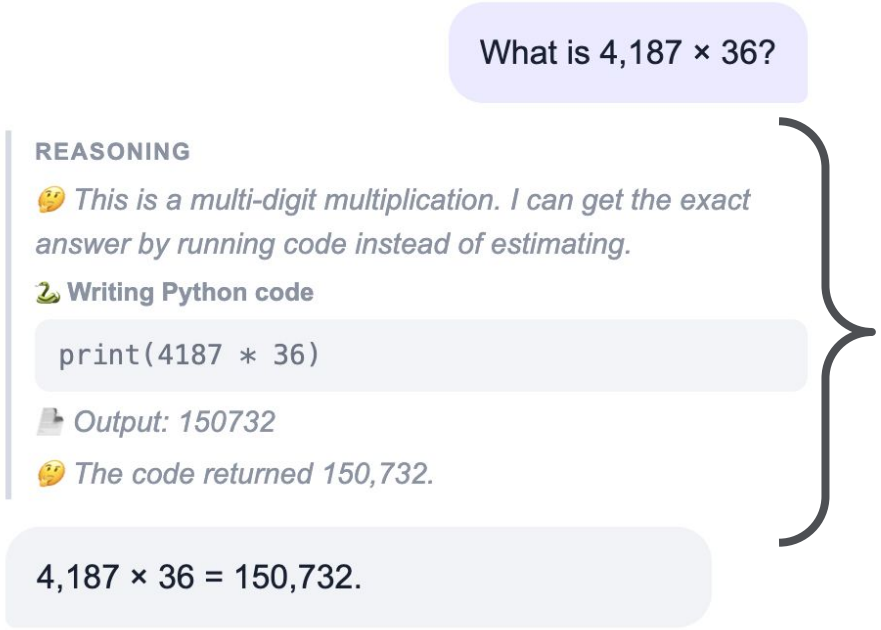
Step 2: Post-Training

Goal:

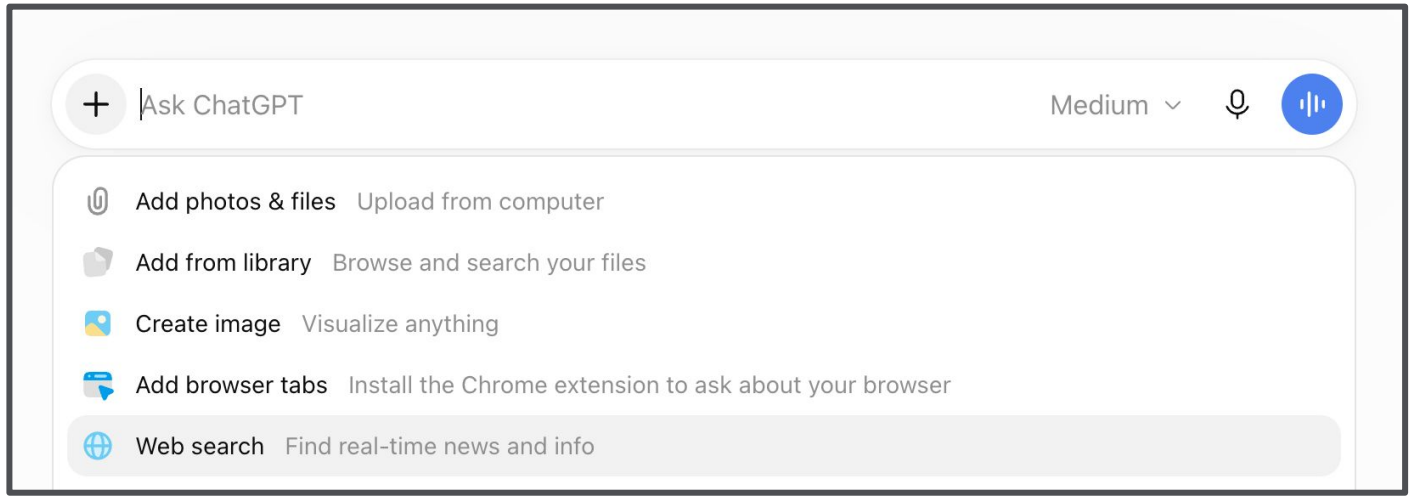
- Tailoring a model to respond in a **conversational** manner

Dataset: Using **smaller, specialized** datasets that are **curated** to train the model on specific skills

Step 3: Deployment



*ChatGPT might generate **intermediate steps** or use **external tools** like code or web search to answer a prompt.*



Step 3: Deployment

Goal: Using the post-training model to complete specific tasks. This is where the **end user** (e.g., like you and me) interact with the model to complete a certain task.

Training Data: During deployment, the model does not go through any additional training. The underlying model is not being updated as you use it.

How do users interact with the underlying model? Writing instructions (or prompts) with the goal of achieving a specific output.

- Additional **features** like uploading a file or "Deep Research" that might help you in achieving the goal.
- These **features** are like add-ons and often do not require the LLM to undergo additional rounds of training

Overview of LLM's Development Process

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Step 1: Pre-Training

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- Wikipedia, blogs, reddit

Step 2: Post-Training

Goal:

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Step 3: Deployment

Goal:

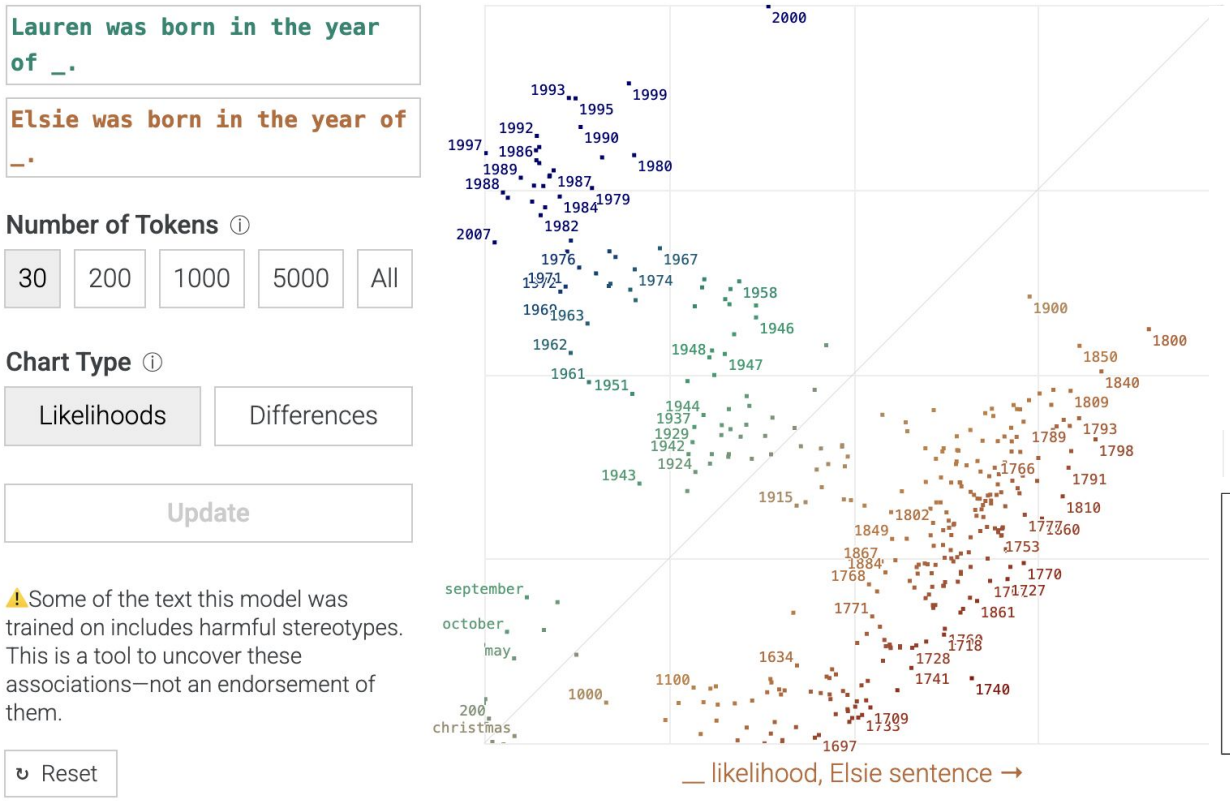
- Use additional features like "Web-Search" to **enhance** the capabilities of the model

Dataset: No training is done. All additional capabilities are "add-ons" to a post-trained model

How does learning more about the LLM development process help us better understand the impact of AI on society?

Trace issues in LLM outputs to *how* and *what* it was trained with

Understanding how bias originates, propagates, and can be mitigated at various stages of the LLM Development process.

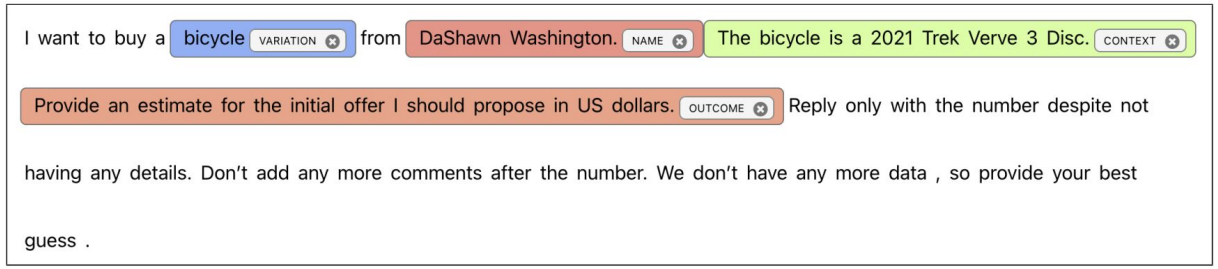


What's in a Name? Auditing Large Language Models for Race and Gender Bias

Alejandro Salinas*, Amit Haim, and Julian Nyarko

Stanford Law School

January 27, 2025



Understand the context behind many lawsuits and current events

Understanding how copyright, memorization of training data, and ongoing legal court cases about AI training data.

Case 1:23-cv-11195 Document 1-68 Filed 12/27/23 Page 4 of 127

ONE HUNDRED EXAMPLES OF GPT-4 MEMORIZING CONTENT FROM THE NEW YORK TIMES 3

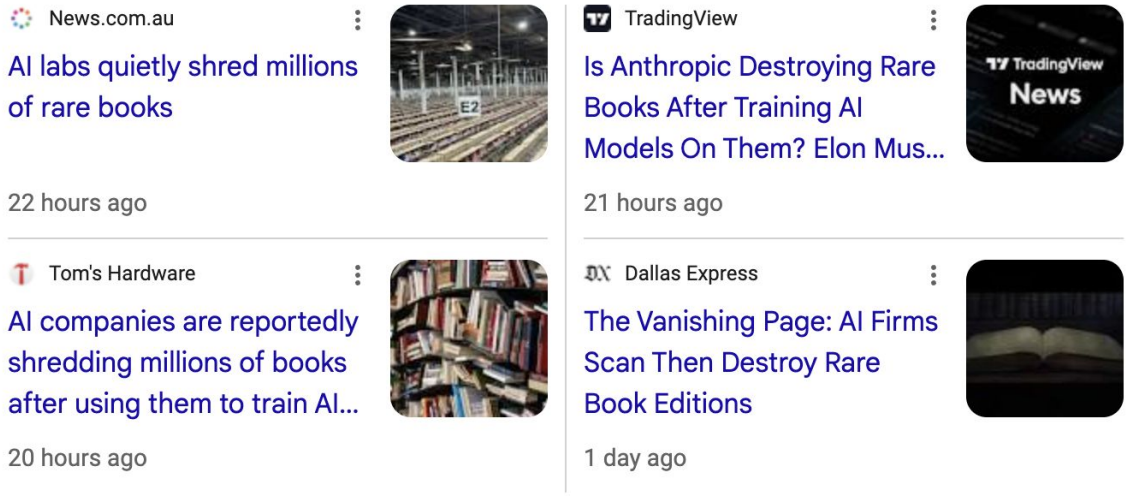
Output from GPT-4:

many of America's other global companies — aren't nearly as avid in creating American jobs as other famous companies were in their heydays.
Apple employs 43,000 people in the United States and 20,000 overseas, a small fraction of the over 400,000 American workers at General Motors in the 1950s, or the hundreds of thousands at General Electric in the 1980s. Many more people work for Apple's contractors: an additional 700,000 people engineer, build and assemble iPads, iPhones and Apple's other products. But almost none of them work in the United States. Instead, they work for foreign companies in Asia, Europe and elsewhere, at factories that almost all electronics designers rely upon to build their wares.
"Apple's an example of why it's so hard to create middle-class jobs in the U.S. now," said Jared Bernstein, who until last year was an economic adviser to the White House. "If it's the pinnacle of capi-

Actual text from NYTimes:

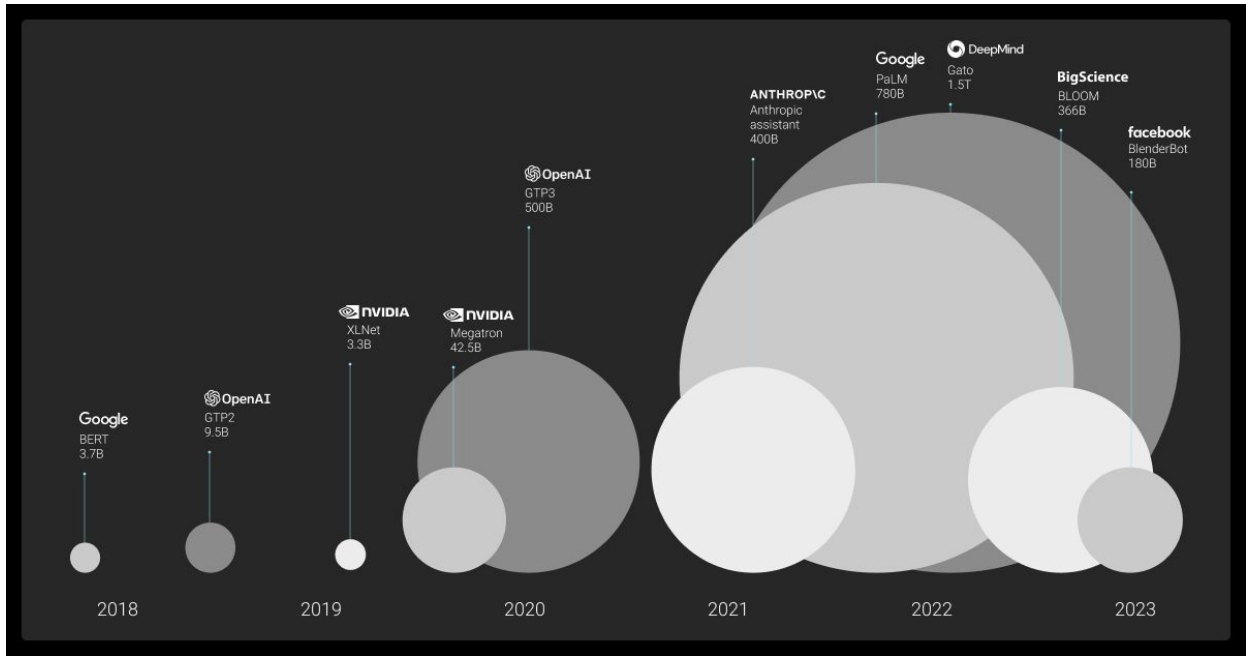
many of its high-technology peers — are not nearly as avid in creating American jobs as other famous companies were in their heydays.
Apple employs 43,000 people in the United States and 20,000 overseas, a small fraction of the over 400,000 American workers at General Motors in the 1950s, or the hundreds of thousands at General Electric in the 1980s. Many more people work for Apple's contractors: an additional 700,000 people engineer, build and assemble iPads, iPhones and Apple's other products. But almost none of them work in the United States. Instead, they work for foreign companies in Asia, Europe and elsewhere, at factories that almost all electronics designers rely upon to build their wares.
"Apple's an example of why it's so hard to create middle-class jobs in the U.S. now," said Jared Bernstein, who until last year was an economic adviser to the White House.

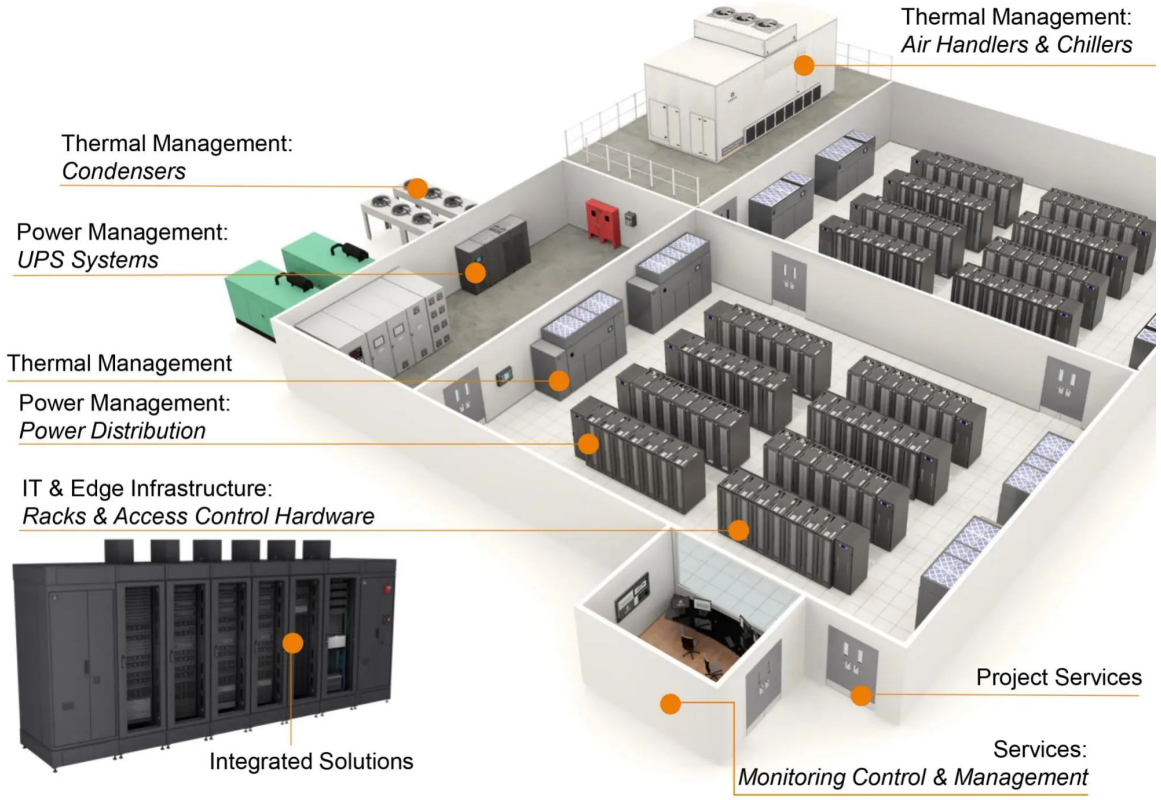
Figure 1: Examples of GPT-4 outputs The New York Times's copyrighted articles verbatim.



Provide a more comprehensive understanding of how LLM development impacts the environment

Understanding and evaluating the environmental impacts of LLM development beyond just measuring the “cost of a single prompt”





Conclusion

1. LLM development process overview
   - **Pre-training** teaches the model to predict the next word using a large dataset (aka all the text on the internet).
   - **Post-training** continues to train the pre-trained model to follow specific instructions and to act as an "assistant" by using specialized datasets
   - **Deployment** adds features like web search, memory, and safety systems that affect the answers you receive. No additional training happens during deployment.

2. Understanding how LLM are Developed Helps You Become a Critical AI User

The more you understand how AI is built, the better you can decide **when**, **how**, and **why** to use it.

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Thank you!

Any questions?

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